PENGUAT BERTINGKAT KUPLING C

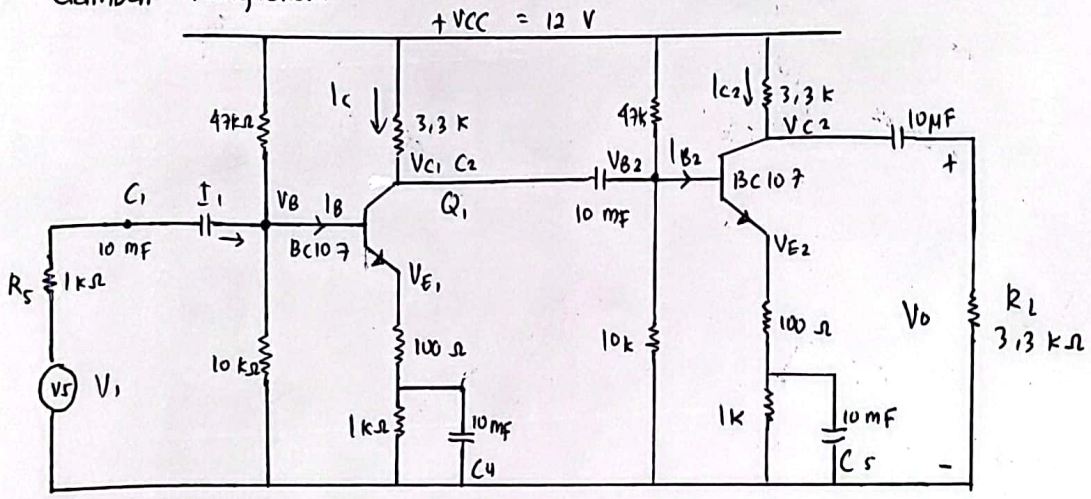
Tanggal Praktikum
31 Agustus 2023

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Kelas : 1A - TC

A. Gambar Rangkaian



B. Tabel Pengukuran

f(Hz)	Vi(mV)	V1	V2	A1	A2	A	
20	10						
200	10						
500	10						
1k	10						
2k	10						
5k	10						
10k	10						
20k	10						
50k	10						
100k	10						
500k	10						
1m	10						
2M	10						

	Ukur	Hitung
V_{B1}	2,06 V	2,1 V
V_{B2}	2,08 V	2,1 V
V_{C1}	7,4 V	7,38 V
V_{C2}	7,5 V	7,38 V
V_{E1}	1,341 V	1,405 V
V_{E2}	1,388 V	1,405 V
I_{C1}	1,48 mA	1,4 mA
I_{C2}	1,4 mA	1,4 mA
I_{E1}	1,41 mA	1,4 mA
I_{e2}	1,45 mA	1,4 mA

C. Hasil Perhitungan

- $R_{th} = \frac{R_1 . R_2}{R_1 + R_2} = \frac{47k . 10k}{47k + 10k} = 8,245\ k\Omega$
- $V_{th} = \frac{10k}{57k} . 12 = 2,105\ V$
- $V_B = V_{Th} = 2,105\ V$
- $V_{CE} = V_{CC} - I_C (R_C + R_E)$
  $= 12 - (1,4)(3,56)$
  $= 7,014\ V$
- $V_E = V_B - V_{BE}$
  $= 2,105 - 0,7$
  $= 1,405\ V$
- $V_C = V_{CC} - I_C . R_C$
  $= 12 - (1,4 . 3,3k)$
  $= 7,38\ V$
- $I_E = \frac{V_E}{R_E} = \frac{1,405}{100} = 1,4\ mA$
- $I_C \cong I_E = 1,4\ mA$

